Sapthagiri College of Engineering

(Affiliated to Visvesvaraya Technological University, Belagavi& Approved by AICTE, New Delhi)

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Department Of Mathematics

Question Bank: Module-3

Sub: MATHEMATICS-I **Sub Code: BMATE/S/C/M101**

1. Solve: $rsin\theta - cos\theta \frac{dr}{d\theta} = r^2$.
2. Solve: $xy(1 + xy^2)\frac{dy}{dx} = 1$.
3. Solve: $x^3 \frac{dy}{dx} - x^2 y = -y^4 \cos x$.
4. Solve: $\left[y\left(1 + \frac{1}{x} + \cos y\right)\right] dx + [x + logx - x\sin y]dy = 0$.
5. Solve: $\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$.
6. Solve: $\left(1 + e^{x/y}\right)dx + e^{x/y}\left(1 - \frac{x}{y}\right)dy = 0$.
7. Solve: $(4xy + 3y^2 - x)dx + x(x + 2y)dy = 0$.
8. Solve: $(x^2 + y^2 + x)dx + xydy = 0$.
9. Find Orthogonal Trajectories of the family of Parabolas $y^2 = 4ax$.
10. Find Orthogonal Trajectories of the family of curves $\frac{x^2}{a^2} + \frac{y^2}{b^2+\lambda} = 1$, Where λ is a parameter.
11. Show that the family of parabolas $y^2 = 4a(x + a)$ is self orthogonal.
12. Find Orthogonal Trajectories of the family $r^n \cos n\theta = a^n$.
13. Find Orthogonal Trajectories of the family of curves $r^n = a^n \cos n\theta$.
14. Find Orthogonal Trajectories of $r^n = a^n \sin n\theta$.
15. Solve: $y\left(\frac{dy}{dx}\right)^2 + (x - y)\frac{dy}{dx} - x = 0$.
16. Solve: $xy\left(\frac{dy}{dx}\right)^2 - (x^2 + y^2)\frac{dy}{dx} + xy = 0$.
17. Solve: $x^2p^2 + xp - (y^2 + y) = 0$.
18. Solve: $\frac{dy}{dx} - \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}$.

19. Modify the equation into Clairaut's form. Hence obtain the associated general and singular solutions. $xp^2 - py + kp + a = 0$.
20. Solve the equation $(px - y)(py + x) = 2p$ by reducing into Clairaut's form, taking the substitution $X = x^2, Y = y^2$.
21. Find the general and singular solution for the equation $x^2(y - px) = p^2 y$ by reducing to Clairaut's form by take the substitution $X = x^2, Y = y^2$.